

Selector Type	Description	XPath Syntax	XPath Example	CSS Selector Syntax	CSS Selector Example
Element Selector	Selects elements based on their tag name (e.g., div, p).	/element	/html/body/div	element	div, p
ID Selector	Selects an element with a specific id attribute.	//*[@id='id_value']	//*[@id='header']	#id_value	#header
Class Selector	Selects elements with a specific class attribute.	//*[@class='class_value']	//*[@class='btn']	.class_value	.btn
Attribute Selector	Selects elements with a specific attribute (e.g., type, href).	//*[@attribute='value']	//input[@type='text']	[attribute='value']	input[type='text']
Descendant Selector	Selects elements that are descendants (any level of children) of the specified parent element.	//parent/descendant	//div/p	parent descendant	div p
Child Selector	Selects direct child elements of a specified parent element.	/parent/child	/div/p	parent > child	div > p
Universal Selector	Selects all elements within the document.	//*	//*	*	*
Following Sibling Selector	Selects elements that are siblings of the current element, located after it in the DOM.	following-sibling::element	//div/following-sibling::p	element + sibling	div + p

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General Sibling Selector	Selects all sibling elements that follow a specified element.	following-sibling::element	//div/following-sibling::p	element ~ sibling	div ~ p
Text Matching	Selects elements whose text content matches a specific string.	text()='text_value'	//button[text()='Click me']	N/A	N/A
Contains Function	Selects elements whose attributes or text contain a specified substring.	contains(attribute, 'value')	//div[contains(@class, 'header')]	element:contains('text')	N/A
First Child Selector	Selects the first child of the parent element.	position()=1 or :first-child	//div[1]	element:first-child	div:first-child
Last Child Selector	Selects the last child of the parent element.	position()=last() or :last-child	//div[last()]	element:last-child	div:last-child
Nth Child Selector	Selects the nth child of a parent element (starting from 1).	position()=n or :nth-child(n)	//div[position()=2]	element:nth-child(n)	div:nth-child(2)
Nth of Type Selector	Selects the nth child of a specific type (e.g., only <div>).	position()=n or :nth-of-type(n)	//div[position()=2]	element:nth-of-type(n)	div:nth-of-type(2)
Not Selector	Excludes elements that match a specific condition.	not()	//div[not(@class='disabled')]	:not(selector)	p:not(.highlight)

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Relative XPath	Selects elements anywhere in the document, relative to the root.	//element	//button	N/A	N/A
Parent Selector	Selects the parent element of the current element.	parent::element	parent::div	N/A	N/A
Wildcard (any element)	Selects any element without specifying its tag.	//* or //element[*]	//*[@id='header']	N/A	N/A
Wildcard (any attribute)	Selects elements with any attribute (not specific to one attribute).	//*[@*='value']	//*[@*='button']	N/A	N/A

Resources

[XPath and Css Selector Tester Tool for Automation Testing Practice](#)
[Webdriveruniversity - Selector Playground](#)
[DEMOQA](#)
[XPath Selector Cheat Sheet: Practical Examples Included](#)

Category	Syntax	Description	Example	Concept Clear
Basic Syntax	/	Selects from the root node	/html selects the root <html> element.	--
	//	Selects nodes from the current node that match, no matter their location	//div selects all <div> elements.	
	.	Selects the current node	. selects the current <div> element in a loop.	
	..	Selects the parent of the current node	../.. selects the parent of the current node.	
	@	Selects attributes	//@href selects all href attributes.	
	nodename	Selects all nodes with the name nodename	//p selects all <p> elements.	
Predicates	[n]	Selects the nth element (1-based index)	//li[3] selects the third element.	
	[position()=n]	Same as [n]	//ul/li[position()=2] selects the second in each .	
	[last()]	Selects the last element	//li[last()] selects the last element.	
	[@attribute='value']	Selects elements with the specified attribute value	//*[@id='unique'] selects elements with id="unique".	
	[contains(@attribute, 'value')]	Selects elements where the attribute contains value	//div[contains(@class, 'note')].	
	[not(predicate)]	Selects elements excluding the predicate	//input[not(@type='hidden')].	
	[starts-with(@attribute, 'value')]	Selects elements where the attribute starts with value	//a[starts-with(@href, 'http')].	
Axes	ancestor::	Selects all ancestors (parent, grandparent, etc.)	//span/ancestor::div.	
	descendant::	Selects all descendants (children, grandchildren, etc.)	//div[@id='content']/descendant::p.	
	following::	Selects all nodes after the current node	//h2/following::p.	
	preceding::	Selects all nodes before the current node	//h2/preceding::p.	
	following-sibling::	Selects siblings after the current node	//h2/following-sibling::div.	
	preceding-sibling::	Selects siblings before the current node	//h2/preceding-sibling::div.	

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	child::	Selects direct children of the current node	//div/child::p.	
	parent::	Selects the parent of the current node	//span/parent::div.	
Wildcards	*	Matches any element node	//* selects all elements.	
	@*	Matches any attribute node	//@* selects all attributes.	
	node()	Matches any node (element, text, etc.)	//body/node().	
Functions	text()	Selects the text content of nodes	//p[text()='Hello World'].	
	contains()	Returns true if the first argument string contains the second argument string	//div[contains(@class, 'important')].	
	starts-with()	Returns true if the first argument string starts with the second argument string	//div[starts-with(@id, 'prefix-')].	
	not()	Returns true if the argument is false	//input[not(@type='hidden')].	
	normalize-space()	Removes extra whitespace	//td[normalize-space(text())='Some text'].	
	translate()	Replaces characters in a string	//text()[translate(., 'ABC', 'abc')='abc'].	
	last()	Returns the position of the last node	//ul/li[last()].	
	position()	Returns the position of the current node	//ul/li[position()<=3].	
	count()	Counts the number of nodes	//ul[count(li)>3].	
	sum()	Returns the sum of the values	sum(//input[@type='number']/@value).	
	floor(), ceil(), round()	Numeric functions	//div[floor(@data-number)=10].	
	By Tag Name	//tagname	Selects all nodes with the name tagname.	
	By Attribute	//*[@attribute='value']	Selects elements with the specified attribute value.	
	By Partial Attribute	//*[contains(@attribute, 'value')]	Selects elements containing value in the specified attribute.	